MPG No. 26.4.98 $\bar{6}'m2'$ (-6'm2' setting) [Type III, hexagonal] [T tensor]

* Rank 0 tensor. * Rank 1 tensor.

$$\begin{bmatrix} 0 & 0 & T_z \end{bmatrix}$$

$$T_z = \mathbb{T}_z^{(1)}$$

* Rank 2 tensor (s). * Rank 2 tensor (a). * Rank 3 tensor (ss).

$$\begin{bmatrix} 0 & 0 & T_{xxz} \\ 0 & 0 & T_{xxz} \\ 0 & 0 & T_{zzz} \\ 0 & T_{xxz} & 0 \\ T_{xxz} & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$T_{xxz} = \mathbb{T}_z^{(1)} - \mathbb{T}_z^{\alpha(1)}$$
$$T_{zzz} = 3\mathbb{T}_z^{(1)} + 2\mathbb{T}_z^{\alpha(1)}$$

* Rank 3 tensor (sa).

$$\begin{bmatrix} 0 & 0 & T_{xxz} \\ 0 & 0 & T_{xxz} \\ 0 & 0 & 0 \\ 0 & -\frac{T_{xxz}}{2} & 0 \\ -\frac{T_{xxz}}{2} & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$T_{xxz} = 2\mathbb{T}_z^{(2)}$$
$$\frac{T_{xxz}}{2} = \mathbb{T}_z^{(2)}$$

* Rank 3 tensor (a).

$$\begin{bmatrix} 0 & T_{yzy} & 0 \\ -T_{yzy} & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$T_{yzy} = \mathbb{T}_z^{(3)}$$

* Rank 4 tensor (sss).

$$\begin{bmatrix} 0 & 0 & 0 & T_{xxyz} & 0 & 0 \\ 0 & 0 & 0 & -T_{xxyz} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ T_{xxyz} & -T_{xxyz} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & T_{xxyz} \\ 0 & 0 & 0 & 0 & T_{xxyz} & 0 \end{bmatrix}$$

$$T_{xxyz} = \mathbb{T}_{4s}^{(1)}$$

* Rank 4 tensor (ssa).

$$\begin{bmatrix} 0 & 0 & 0 & T_{xxyz} & 0 & 0 \\ 0 & 0 & 0 & -T_{xxyz} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ -T_{xxyz} & T_{xxyz} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -T_{xxyz} \\ 0 & 0 & 0 & 0 & T_{xxyz} & 0 \end{bmatrix}$$

$$T_{xxyz} = 2\mathbb{M}_{3c}^{(1)}$$

* Rank 4 tensor (aas). * Rank 4 tensor (aaa). * Rank 4 tensor (sa).

$$\begin{bmatrix} T_{xxyz} & 0 & 0 \\ -T_{xxyz} & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & -T_{xxyz} & 0 \end{bmatrix}$$

$$T_{xxyz} = \mathbb{M}_{3c}^{(2)}$$

* Rank 4 tensor (as).

$$\begin{bmatrix} T_{yzxx} & -T_{yzxx} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -T_{yzxx} \\ 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

$$T_{yzxx} = \mathbb{M}_{3c}^{(3)}$$

* Rank 4 tensor (s).

$$\begin{bmatrix} 0 & 0 & 0 & T_{xxyz} & 0 & 0 & T_{xxzy} & 0 & 0 \\ 0 & 0 & 0 & -T_{xxyz} & 0 & 0 & -T_{xxzy} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ T_{yzxx} & -T_{yzxx} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & T_{yzxx} & 0 & 0 & T_{yzxx} \\ 0 & 0 & 0 & 0 & T_{xxzy} & 0 & 0 & T_{xxyz} & 0 \end{bmatrix}$$

$$T_{xxyz} = 2\mathbb{M}_{3c}^{(1)} + \mathbb{M}_{3c}^{(2)} + \mathbb{T}_{4s}^{(1)}$$
$$T_{xxzy} = 2\mathbb{M}_{3c}^{(1)} - \mathbb{M}_{3c}^{(2)} + \mathbb{T}_{4s}^{(1)}$$
$$T_{yzxx} = -2\mathbb{M}_{3c}^{(1)} + \mathbb{T}_{4s}^{(1)}$$

* Rank 4 tensor (a).

$$\begin{bmatrix} T_{yzxx} & -T_{yzxx} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -T_{yzxx} & 0 & 0 & -T_{yzxx} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

$$T_{yzxx} = \mathbb{M}_{3c}^{(3)}$$

* Rank 4 tensor (t).

$$\begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & T_{yyyz} \\ 0 & 0 & 0 \\ 0 & T_{yyyz} & 0 \\ 0 & 0 & 0 \\ 0 & 0 & -T_{yyyz} \\ 0 & 0 & 0 \\ 0 & -T_{yyyz} & 0 \\ 0 & 0 & 0 \\ -T_{yyyz} & 0 & 0 \end{bmatrix}$$

$$T_{yyyz} = -\mathbb{T}_{4s}^{(1)}$$